

## CO3\_AT1\_DEBUGGING

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## CSA0927-JAVA PROGRAMMING

1.

### Q1 Debug Nested Exception Handling

10 marks

```
import java.util.Scanner;

public class Debug1 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = {20,40,60,80};

        int index = sc.nextInt();
        int divisor = sc.nextInt();

        try {
            int value = arr[index];
            try {
                int result = value/divisor;
                System.out.println(result);
            }
            catch(ArrayIndexOutOfBoundsException e) {
                System.out.println("Invalid Index");
            }
        }
        catch(ArithmeticException e) {
            System.out.println("Division by Zero");
        }
    }
}
```

Debugging Task

Identify why some exceptions are never caught.

Modify the exception handling so all runtime exceptions are handled correctly.

TEST CASE 1

Input 2 5

Expected Output: 12

TEST CASE 2

Input 5 2

Expected Output : Invalid Index

TEST CASE 3

Input 1 0

Expected Output : Division by Zero

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

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Your Fixed Code

```
1 import java.util.Scanner;
2 public class Debug1{
3     public static void main(String args[]){
4         Scanner sc = new Scanner (System.in);
5         int arr[]={20,40,60,80};
6         int index = sc.nextInt();
7         int divisor = sc.nextInt();
8         try {
9             int value = arr [index];
10            int result = value / divisor;
11            System.out.println(result);
12        } catch (ArrayIndexOutOfBoundsException e){
13            System.out.println("Invalid Index");
14        } catch (ArithmeticException e){
15            System.out.println("Division by Zero");
16        }
17    }
18 }
```

Test Input

1 0

Output

Division by Zero

Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

2.

Q2 Debug Exception Propagation

10 marks

```

import java.util.Scanner;

class Calculator{
    static int divide(int a,int b){
        return a/b;
    }

    static void display(int x){
        System.out.println("Answer = "+x);
    }
}

public class Debug2{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);

        int a=sc.nextInt();
        int b=sc.nextInt();

        Calculator.display(Calculator.divide(a,b));

        System.out.println("Execution Completed");
    }
}

Debugging Task
Prevent abnormal termination.
Ensure "Execution Completed" is always printed.
TEST CASE 1
INPUT: 20 5
OUTPUT: Answer = 4
Execution Completed
TEST CASE 2
INPUT: 50 0
OUTPUT: Cannot Divide by Zero
Execution Completed
TEST CASE 3
INPUT: 36 6
OUTPUT: Answer = 6
Execution Completed

```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

#### Your Fixed Code

```

1 import java.util.Scanner;
2 public class Calculator {
3     static int divide(int a, int b) {
4         try {
5             return a / b;
6         } catch (ArithmeticException e) {
7             System.out.println("Cannot Divide by Zero");
8             return -1;
9         }
10    }
11    static void display(int x) {
12        if (x != -1) {
13            System.out.println("Answer = " + x);
14        }
15    }
16    public static void main(String args[]) {
17        Scanner sc = new Scanner(System.in);
18        int a = sc.nextInt();
19        int b = sc.nextInt();
20        display(divide(a, b));
21        System.out.println("Execution Completed");
22    }
23 }

```

✓ Submitted

#### Test Input

20 5

#### Output

```

Answer = 4
Execution Completed

```

#### Test Cases

|             |   |
|-------------|---|
| Test Case 1 | ✓ |
| Test Case 2 | ✓ |
| Test Case 3 | ✓ |

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3.

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Q3 Debug Multiple Runtime Exceptions10 marks

```
import java.util.Scanner;

public class Debug3 {

    public static void main(String args[]) {

        Scanner sc=new Scanner(System.in);

        int arr[]={12,24,36,48};

        String number=sc.next();

        int index=Integer.parseInt(number);

        System.out.println(arr[index]);

    }

}
```

Debugging Task

The program may generate two different runtime exceptions.

Modify the program so that each exception is handled separately.

Test Case 1

INPUT: 2

OUTPUT: 36

Test Case 2

INPUT: ABC

OUTPUT: Invalid Number

Test Case 3

INPUT: 5

OUTPUT: Invalid Array Index

## Your Fixed Code

```
1 import java.util.Scanner;
2
3 public class Debug3 {
4     public static void main(String args[]) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int arr[] = {12, 24, 36, 48};
9
10        String number = sc.next();
11
12        try {
13            int index = Integer.parseInt(number);
14
15            try {
16                System.out.println(arr[index]);
17            } catch (ArrayIndexOutOfBoundsException e) {
18                System.out.println("Invalid Array Index");
19            }
20
21        } catch (NumberFormatException e) {
22            System.out.println("Invalid Number");
23        }
24
25        sc.close();
26    }
27 }
```

 Submitted

## Test Input

12

## Output

Invalid Array Index

## Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

4.

#### Q4 Debug Finally Execution

10 marks

```
public class Debug4 {
    static int calculate(int a,int b){
        try{
            return a/b;
        }
        finally{
            System.out.println("Resources Released");
        }
    }

    public static void main(String args[]){
        System.out.println(calculate(20,0));
    }
}
```

##### Debugging Task

Prevent program termination.

Ensure the finally block executes while handling the exception correctly.

Test Case 1

INPUT: 20 5

OUTPUT: Resources Released

4

Test Case 2

20 0

Resources Released

Division by Zero

Test Case 3

81 9

Resources Released

9

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

#### Your Fixed Code

```
1 import java.util.Scanner;
2
3 public class Debug4 {
4
5     static int calculate(int a, int b) {
6         try {
7             return a / b;
8         } finally {
9             System.out.println("Resources Released");
10        }
11    }
12
13    public static void main(String args[]) {
14        Scanner sc = new Scanner(System.in);
15        int a = sc.nextInt();
16        int b = sc.nextInt();
17
18        try {
19            int result = calculate(a, b);
20            System.out.println(result);
21        } catch (ArithmeticException e) {
22            System.out.println("Division by Zero");
23        }
24    }
25 }
```

✓ Submitted

#### Test Input

20 5

#### Output

```
Resources Released
4
```

#### Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

5.

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Q5 Debug the NullPointerException

10 marks

```
public class Debug4 {
    public static void main(String[] args) {
        String name = null;
        System.out.println(name.length());
    }
}
```

Expected Task  
Prevent the program from crashing.

Test Case 1  
INPUT: Java  
OUTPUT: Length = 4

Test Case 2  
INPUT: null  
OUTPUT: String is Null

Test Case 3  
INPUT: Programming  
OUTPUT: Length = 11

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

## Your Fixed Code

```
1 import java.util.Scanner;
2
3 public class Debug4 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String name = sc.nextLine();
7
8         if (name == null || name.equals("null")) {
9             System.out.println("String is Null");
10        } else {
11            System.out.println("Length = " + name.length());
12        }
13    }
14 }
```

## Test Input

null

## Output

String is Null

## Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

Submitted